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Towards weeds identification assistance through transfer learning

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Highlights

- Weed identification using deep learning techniques.
- Different pre-processing and transfer learning strategies were evaluated.
- The performance of 7 state-of-art deep neural architectures were performed.
- A dataset containing annotated crops and weeds has been published.
- The methodology proposed can be used by either RGB or multispectral pictures.

Abstract

Reducing the use of pesticides through selective spraying is an important component towards a more sustainable computer-assisted agriculture. Weed identification at early growth stage contributes to reduced herbicide rates. However, while computer vision alongside deep learning have overcome the performance of approaches that use hand-crafted features, there are still some open challenges in the development of a reliable automatic plant identification system. These type of systems have to take into account different sources of variability, such as growth stages and soil conditions, with the added constraint of the limited size of usual datasets. This study proposes a novel crop/weed identification system that relies on a combination of fine-tuning pre-trained convolutional networks (Xception, Inception-Resnet, VGNets, Mobilenet and

Densenet) with the “traditional” machine learning classifiers (Support Vector Machines, XGBoost and Logistic Regression) trained with the previously deep extracted features. The aim of this approach was to avoid overfitting and to obtain a robust and consistent performance. To evaluate this approach, an open access dataset of two crop [tomato (*Solanum lycopersicum* L.) and cotton (*Gossypium hirsutum* L.)] and two weed species [black nightshade (*Solanum nigrum* L.) and velvetleaf (*Abutilon theophrasti* Medik.)] was generated. The pictures were taken by different production sites across Greece under natural variable light conditions from RGB cameras. The results revealed that a combination of fine-tuned Densenet and Support Vector Machine achieved a micro F_1 score of 99.29% with a very low performance difference between train and test sets. Other evaluated approaches also obtained repeatedly more than 95% F_1 score. Additionally, our results analysis provides some heuristics for designing transfer-learning based systems to avoid overfitting without decreasing performance.

Keywords

Weed identification; Deep learning; Transfer learning; Open data; Precision agriculture

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Introduction

Since the world population is growing continuously, the production of high nutritional food must be increased in the next years (Kitzes et al., 2008, United Nations (UN), 2019), accompanied with the protection of natural ecosystems by using sustainable farming procedures (Food and Agriculture Organization, 2013). Among these actions, one of the most challenging is an effective reduction of the impact caused by weeds. In recent years, their damage, alongside with other pests, accounts for about 40% of global yield losses and is expected to increase in the coming years (European Crop Protection Association (ECPA), 2017). Weeds compete with crop plants for resources such as water, nutrients, light and space, causing crop yield losses. Additionally, weeds reduce the quality of farm products, cause irrigation water loss and hinder the operation of harvesting machines and, consequently, decrease the commercial value of cultivated areas (Rizzardi and Fleck, 2016, Zimdahl, 2018). To deal with weeds, farmers tend to apply uniform herbicide spraying throughout the field, usually two or three times in each growing season. However, this practice has as a result to apply uncontrolled large quantities of herbicides, which is harmful for humans, non-target organisms and the environment (Oerke, 2006, Zimdahl, 2018).

Currently, a key objective is the deployment of new information and communication technologies solutions to reduce the reliance on agrochemical products and to reduce human errors caused by cognitive phenomena (Bock et al., 2010). Specifically, the combination of smartphones with high performance processors, HD cameras, sensors, and cloud computing capabilities allow weed detection, when suitable human assessment is unavailable. Consequently, a most effective weed management is possible, contributing to the improvement of farm productivity, food security and environmental sustainability (Gebbers and Adamchuk, 2010).

Various studies have been carried out aiming on an accurate, reproducible and automatic identification method for weeds. The majority of these studies were mostly concerned with acquisition, preprocessing, extraction of manually-designed features and supervised classifiers (Haug et al., 2014, Lottes et al., 2016, Kounalakis et al., 2016, Kounalakis et al., 2017, Lottes et al., 2017, Bakhshipour and Jafari, 2018). Classifiers used for this purpose as Support Vector Machines (SVM) and traditional/shallow neural networks are dependent on good feature extractors, such as Scale-Invariant Feature Transform (SIFT). However, the main limiting factor for these weed recognition systems is the limited ability of their manually-designed extracted features (colour information, shape analysis or texture analysis) to effectively represent image content. This factor adds up to a variety of challenges associated with image classification, such as viewpoint, scale and intra-class variations, image deformation, image occlusion, illumination conditions, and background clutter.

Furthermore, most previous studies based on hand-crafted features require tedious operations in the case of extensive preprocessing, image normalization and plant segmentation, which may significantly restrict the repeatability of the identification method. In recent years, researchers on weed or other plant identification developed systems incorporating deep learning methodologies. According to Ferentinos, (2018), deep learning refers to “the use of artificial neural network, architectures that contain a quite large cascade of multiple processing layers, as opposed to shallower architectures of more traditional neural network methodologies”. With the inception of deep learning concepts, the solution to the limitations of the hand-crafted approaches seems to be closer than ever, as a result of the automatic feature extraction from the raw data (Lecun et al., 2015). Among the deep learning tools for weed identification, the most commonly used are the Convolutional Neural Networks (CNN) (Krizhevsky et al., 2012). This kind of neural network is complex but efficient, with a high rate of discrimination and have proven to provide good results in precision agriculture for the correct identification of plants. Compared to previous methods, the self-learned features make the CNN less affected by natural variations such as changes in illumination, skewed leaves and occluded plants. Currently, CNNs are used in end-to-end crop/weed identification systems to overcome the limitations of hand-crafted approaches and reaching state-of-the-art performance. Potena et al., (2016) used a cascade of CNNs for crop/weed identification, where the first CNN detected vegetation and then the vegetation pixels were classified by a deeper crop-weed CNN. They used a multispectral camera to obtain RGB and NIR images. McCool et al., (2017) fine-tuned a very deep CNN and gained practical processing times by compression of the fine-tuned network using a mixture of small, but fast networks, without losing too much identification accuracy. Fully CNNs directly estimate a pixel-wise segmentation of the complete image and can use information from the whole image. Finally, Milioto et al., (2018) proposed a deep encoder-decoder CNN that exploits existing vegetation indexes and provides an identification in real-time.

Another important breakthrough has been the use of transfer learning in order to relax the data requirement of the CNNs (Ferentinos, 2018). Transfer learning recycles previously trained networks by using the new data to update a small part of the original weights (Bengio, 2012). The new features learned can prove useful for many different problems, even though these new problems may involve completely different classes than those of the original task. Wang et al., (2017) used transfer learning in order to obtain the best neural-based method for disease detection in plants. Our work is highly related with Suh et al., (2018), where the authors deeply compared different transfer learning approaches in order to find a suitable approach for weed detection (volunteer potato). Finally, another relevant study has been performed by Kounalakis et al., (2019) where they evaluated transfer learning by a combination of CNN-based feature extraction and linear classifiers to recognize rumex under real-world conditions.

In this paper, we present a novel methodology to find a suitable neural network that automatically performs the crop/weed identification tasks in real-time. To face this fine-grained image classification, we used the combination of fine-tuned deep neural networks for feature extraction with other machine learning classifiers on the top of them. To explore the best architecture and training mechanism, we fine-tuned some of the state-of-the-art deep neural networks. Our approach exploits a pipeline that includes different CNNs applied to the input images, where we can find tomato (*Solanum lycopersicum* L.), cotton (*Gossypium hirsutum* L.) and two weeds in relation to them: the black nightshade (*Solanum nigrum* L.) and velvetleaf (*Abutilon theophrasti* Medik.). Additionally, since there are few public datasets for weeds and multi-spectral cameras are still too expensive for small farmers, we have developed our own dataset to perform the experiments and validate our methodology. We have taken data from different farms in order to confirm the effectiveness of our approach. The pictures were obtained under natural variable light conditions from an RGB camera, which is the main difference from other datasets which use NIR or RGB+NIR cameras (Potena et al., 2016, Haug and Ostermann, 2018). However, it is important to underpin that the approach presented in this paper is also applicable for multi-spectral images.

In summary, our contribution is three-fold. Firstly, we have developed a novel vision-based solution for automated crop/weed identification from unconstrained RGB plant photographs by utilising fine-tuned deep neural networks. Secondly, we made an in-depth performance evaluation of the critical factors affecting the fine-tuning of pre-trained models, such as background removal, pooling type, and strategy for transfer learning. Our findings determine the relative significance of each of the aforementioned variables on performance, thus paving the way for more efficient utilisation of valuable computational resources. Finally, we provide an open dataset with crop/weed RGB images under natural variable light conditions where our methodology and future research could be evaluated.

This article is structured as follows: Section 2 presents our transfer learning approach and the crop/weed dataset. Section 3 explains the experimental setup and results. Section 4 presents a discussion about the results and in Section 5 conclusions are drawn and future directions envisaged.

Section snippets

Material and methods

The primary goal of this research was to develop a combination of pre-trained deep neural networks and a classifier to robustly identify crops and weeds in the field, even when the visual appearance of the crops and soil has changed. As shown in Fig. 1, the approach is composed of different stages. At each stage there are different configurations that were empirically evaluated to find the best crop/weed classifier. The first stage consisted of the capture of crops and weeds images, and their …

Experimental results

This section shows the results of the experiments performed with the different neural architectures in order to find the most suitable approach for identifying weeds. Moreover, our experiments were designed to show the capabilities of our method and to support our key claims, which are: (i) We can design an end-to-end approach for identifying crop/weed with high performance and low train/test difference; (ii) the use of a traditional shallow classifier on the top of deep features can improve …

Discussion

In this work, a crop/weed classifier has been developed, which is an important step to support crop protection and prevent weed dispersal, due to the more effective weed management, when expert knowledge is unavailable. The methodology was composed of several steps which can adopt different configurations. The main objective was to find the configuration that offered the best performance according to our evaluation criteria. In this case, the system "NPS:DenseNet:FT:AMP+SVM" achieved the best …

Conclusions

In this study, a novel vision-based classification system for identifying weeds in crop fields, by exploiting pre-trained deep neural networks, was examined. The study focused on volunteer tomato and volunteer cotton, common weeds for tomato plants and cotton plants, respectively. We have performed an extensive evaluation of the algorithms using real data. The best crop/weed identifier has obtained a promising performance of 99.29% F_1 score besides a low tendency to overfitting. Moreover, …

CRediT authorship contribution statement

Borja Espejo-Garcia: Investigation, Software, Writing - original draft. **Nikos Mylonas:** Data curation, Resources, Writing - original draft. **Loukas Athanasakos:** Data curation, Resources, Writing - original draft. **Spyros Fountas:** Supervision, Writing - review & editing. **Ioannis Vasilakoglou:** Supervision, Writing - review & editing. …

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…As a rough rule of thumb, training deep learning models from scratch requires thousands of images per category for achieving human-level performance (Goodfellow et al., 2016). Transfer learning can reduce the image number requirement to as few as hundreds or even tens of images (Espejo-Garcia et al., 2020; Suh et al., 2018), but it should be noted that such approaches heavily rely on fine turning of the models pre-trained using the images that are generally irrelevant to domain precision agriculture tasks, which hence may not generalize well in practical applications. Despite the recognized need for larger datasets, the collection of sufficiently large datasets can be a daunting task due to the manual efforts and costs involved, and in some cases even infeasible for certain classes (e.g., rare diseases or weed species) that have very few occurrences in the field.…

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